

# **RYGAR TECHNOLOGIES INTERNSHIP**

## **SMART NEONATAL INCUBATOR MONITORING AND TEMPERATURE CONTROL SYSTEM**

**A PROJECT REPORT**

*Submitted by*

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# **SMART NEONATAL INCUBATOR MONITORING AND TEMPERATURE CONTROL SYSTEM**

## **ABSTRACT:**

Premature and newborn babies require a carefully controlled environment to maintain a stable body temperature. Any sudden increase or decrease in incubator temperature may lead to serious health complications. The proposed Smart Neonatal Incubator Monitoring and Temperature Control System is an embedded system designed to monitor incubator temperature continuously and provide automatic control and alerts. The system uses an Arduino Uno, LCD display, cooling fan, and buzzer to maintain a safe environment for infants. Since temperature sensors were difficult to simulate effectively in Tinkercad, a potentiometer was used to emulate different temperature conditions. The system displays the simulated temperature, activates a cooling fan when the temperature exceeds a predefined threshold, and generates an alarm during dangerous conditions. This project demonstrates the application of embedded systems in biomedical engineering and infant healthcare monitoring.

## INTRODUCTION:

Neonatal care plays a vital role in ensuring the survival and healthy development of premature and newborn infants. Premature babies are born before the completion of the normal gestation period and often have underdeveloped organs and physiological systems. One of the major challenges faced by premature infants is the inability to regulate their body temperature effectively. Unlike healthy adults, newborn babies have a larger body surface area relative to their weight, less body fat, and immature temperature regulation mechanisms, making them highly vulnerable to temperature fluctuations.

The normal body temperature of a healthy newborn infant typically ranges between **36.5°C and 37°C**. Maintaining this temperature range is essential for proper metabolic function, growth, and overall health. If the infant's body temperature falls below the normal range (hypothermia), it can lead to breathing difficulties, reduced oxygen supply, low blood sugar levels, and increased risk of infection. Similarly, excessive temperature (hyperthermia) can result in dehydration, increased heart rate, and other serious health complications.

To address these challenges, neonatal incubators are used in hospitals and neonatal intensive care units (NICUs). A neonatal incubator is a specialized medical device designed to provide a controlled environment for newborn babies by maintaining optimal temperature, humidity, and air circulation. The recommended incubator temperature generally ranges from **35°C to 37°C**, depending on the infant's weight, age, and medical condition. Continuous monitoring of incubator temperature is necessary to ensure that the infant remains in a safe and comfortable environment.

Traditional incubators often require periodic manual monitoring by healthcare professionals. Human error, delayed response, or equipment malfunction may result in unsafe temperature conditions, potentially affecting the infant's health. Therefore, there is a need for an automated monitoring system that continuously observes the incubator temperature and provides immediate corrective actions when abnormal conditions occur.

The proposed **Smart Neonatal Incubator Monitoring and Temperature Control System** is an embedded-system-based solution developed to monitor and regulate incubator temperature automatically. The system utilizes an Arduino Uno microcontroller, LCD display, cooling fan, buzzer, and a temperature input source. In the Tinkercad simulation environment, a potentiometer is used to simulate temperature variations because of the limitations associated with temperature sensor simulation. In a real-world implementation, the potentiometer can be replaced with a temperature sensor such as **LM35**, **TMP36**, or **DHT11** to measure actual incubator temperature.

The Arduino continuously reads the temperature input and displays the current status on the LCD screen. When the temperature exceeds the normal range, the cooling fan is automatically activated to reduce the temperature. If the temperature reaches a dangerous level, an audible buzzer alert is generated to notify healthcare personnel. This automation enhances patient

safety, reduces the need for constant manual supervision, and demonstrates the application of embedded systems in biomedical engineering and neonatal healthcare.

This project serves as a prototype for smart healthcare monitoring systems and highlights how embedded technology can be utilized to improve neonatal care, increase reliability, and support healthcare professionals in maintaining optimal conditions for newborn infants.

## **PROBLEM STATEMENT:**

Premature babies are unable to regulate their body temperature efficiently and therefore require neonatal incubators to provide a controlled thermal environment. Manual monitoring of incubator temperature may lead to delayed responses in abnormal situations, which can affect infant health. Hence, there is a need for an automated system that continuously monitors temperature and provides immediate corrective actions and alerts.

## **PROPOSED SOLUTION:**

The proposed system uses an Arduino Uno to monitor the incubator temperature and take appropriate actions based on predefined temperature ranges. The temperature value is displayed on an LCD screen for continuous monitoring. When the temperature exceeds the safe range, a cooling fan is automatically activated. If the temperature reaches a dangerous level, a buzzer alert is generated to notify healthcare personnel. This automation helps maintain a safe environment for neonatal care and reduces the need for continuous manual supervision.

## **COMPONENTS USED:**

| <b>S.No</b> | <b>Component</b>        | <b>Purpose</b>                                             |
|-------------|-------------------------|------------------------------------------------------------|
| 1           | Arduino Uno             | Main controller used to process inputs and control outputs |
| 2           | Potentiometer           | Simulates temperature variations in Tinkercad              |
| 3           | 16×2 LCD Display        | Displays temperature status and system condition           |
| 4           | DC Motor (Fan)          | Provides cooling when temperature exceeds the safe limit   |
| 5           | Buzzer                  | Generates alert during dangerous temperature conditions    |
| 6           | NPN Transistor (2N2222) | Acts as a switch to drive the DC motor                     |
| 7           | Resistors               | Current limiting and circuit protection                    |
| 8           | Jumper Wires            | Electrical connections between components                  |
| 9           | Breadboard (Optional)   | Easy circuit assembly and testing                          |

## COMPONENT DESCRIPTION:

### Arduino Uno:

The Arduino Uno serves as the brain of the system. It reads analog values from the potentiometer and controls the LCD, fan, and buzzer based on programmed conditions.

### Potentiometer:

A potentiometer is used in the simulation to represent different temperature levels. By rotating the knob, different voltage levels are applied to the Arduino analog input.

### LCD Display:

The 16×2 LCD displays the simulated temperature and system status such as Normal, Fan ON, and Danger.

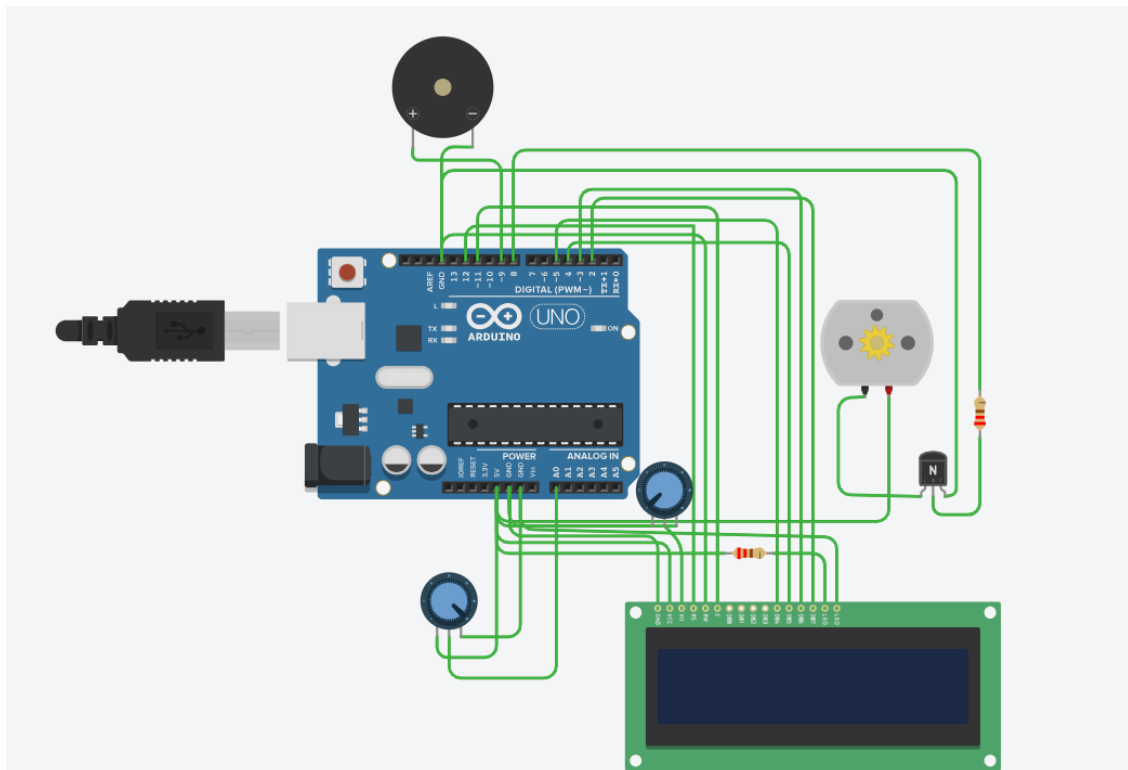
### DC Motor:

The DC motor represents the cooling fan of the incubator. It automatically turns ON when the temperature exceeds the normal range.

### Buzzer:

The buzzer generates an audible alert whenever the incubator reaches a dangerous temperature condition.

## CIRCUIT CONNECTIONS:



### Potentiometer (Temperature Simulation)

| Potentiometer Pin | Connection |
|-------------------|------------|
| Left Pin          | 5V         |
| Middle Pin        | A0         |
| Right Pin         | GND        |

### LCD Connections

| LCD Pin | Arduino Connection     |
|---------|------------------------|
| RS      | D12                    |
| E       | D11                    |
| D4      | D5                     |
| D5      | D4                     |
| D6      | D3                     |
| D7      | D2                     |
| VSS     | GND                    |
| VDD     | 5V                     |
| RW      | GND                    |
| V0      | Contrast Potentiometer |
| LED+    | 5V (through resistor)  |
| LED-    | GND                    |

### Fan (DC Motor) Connections

| Component          | Connection                  |
|--------------------|-----------------------------|
| Motor Positive     | 5V                          |
| Motor Negative     | Collector of NPN Transistor |
| Transistor Base    | Arduino D8 through resistor |
| Transistor Emitter | GND                         |

### Buzzer Connections

| Buzzer Pin        | Connection |
|-------------------|------------|
| Positive Terminal | Arduino D9 |
| Negative Terminal | GND        |

## **WORKING PRINCIPLE:**

The potentiometer is connected to the analog input pin (A0) of the Arduino. The Arduino continuously reads the analog value and interprets it as a simulated temperature. Based on the obtained value:

- Low range: Normal condition
- Medium range: Cooling fan activated
- High range: Cooling fan and buzzer activated

The current status and temperature are displayed on the LCD.

## **TINKERCAD LINK:**

[https://www.tinkercad.com/things/3ZURzxv3Xwo/editel?returnTo=%2Fcircuits&sharecode=r11LVY9v-Vsuj-oizOFK4eO-38hQKgj-C\\_pQDfxKls](https://www.tinkercad.com/things/3ZURzxv3Xwo/editel?returnTo=%2Fcircuits&sharecode=r11LVY9v-Vsuj-oizOFK4eO-38hQKgj-C_pQDfxKls)

## **SYSTEM OPERATION:**

1. The potentiometer simulates temperature changes.
2. Arduino reads the analog value through pin A0.
3. The analog value is converted into a digital value using the internal ADC.
4. Based on predefined thresholds:
  - ADC Value 0–300 → Normal Condition
  - ADC Value 301–700 → Fan Activated
  - ADC Value 701–1023 → Fan and Buzzer Activated
5. The LCD displays the current status of the incubator.
6. The fan and buzzer provide automatic control and warning functions.

## **POTENTIOMETER-BASED TEMPERATURE SIMULATION:**

In a real neonatal incubator, a temperature sensor such as LM35 or TMP36 would be used to measure the actual temperature. However, due to simulation limitations in Tinkercad, a potentiometer was used to simulate temperature changes.

The potentiometer acts as a variable voltage divider. By rotating the knob, different voltages are supplied to the Arduino analog input pin.

## **SIMULATED TEMPERATURE RANGES:**

| <b>ADC Value</b> | <b>Simulated Temperature Condition</b> |
|------------------|----------------------------------------|
| 0 – 300          | Normal (35°C)                          |
| 301 – 700        | High Temperature (38°C)                |
| 701 – 1023       | Dangerous Temperature (40°C)           |

## **ADC (ANALOG-TO-DIGITAL CONVERTER) OPERATION:**

Arduino Uno contains a 10-bit ADC.

- Input Voltage Range: 0V to 5V
- ADC Resolution: 10-bit
- Output Range: 0 to 1023

The ADC converts analog voltages into digital values.

Examples:

- 0V → 0
- 2.5V → 512
- 5V → 1023

The Arduino uses the `analogRead()` function to read these values and determine the corresponding temperature condition.

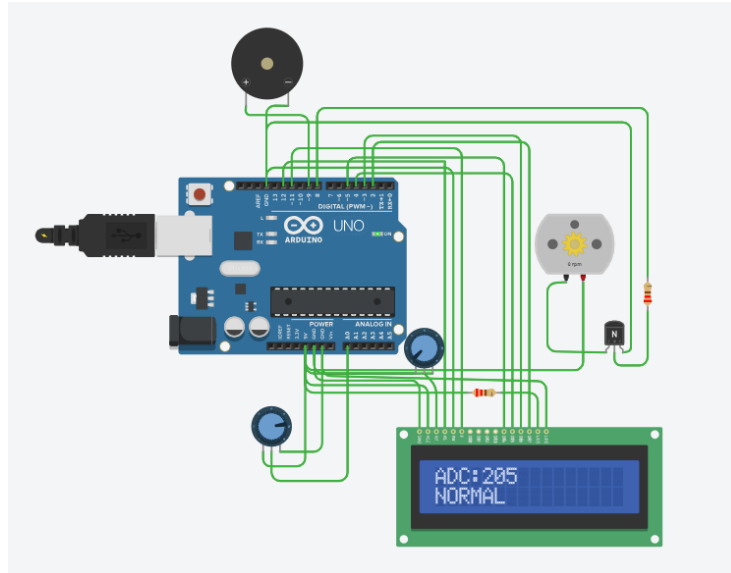
## **RESULTS AND DISCUSSION:**

The Smart Neonatal Incubator Monitoring and Temperature Control System was successfully designed and simulated using Tinkercad. The system demonstrated the ability to monitor simulated temperature conditions and respond appropriately based on predefined threshold values. A potentiometer was used to simulate temperature variations, and the Arduino Uno processed the analog input through its built-in 10-bit ADC.

During the simulation, three different operating conditions were tested:

### **1. Normal Condition**

When the potentiometer produced an ADC value between **0 and 300**, the system interpreted the incubator temperature as approximately **35°C**, which falls within the safe operating range for neonatal care.

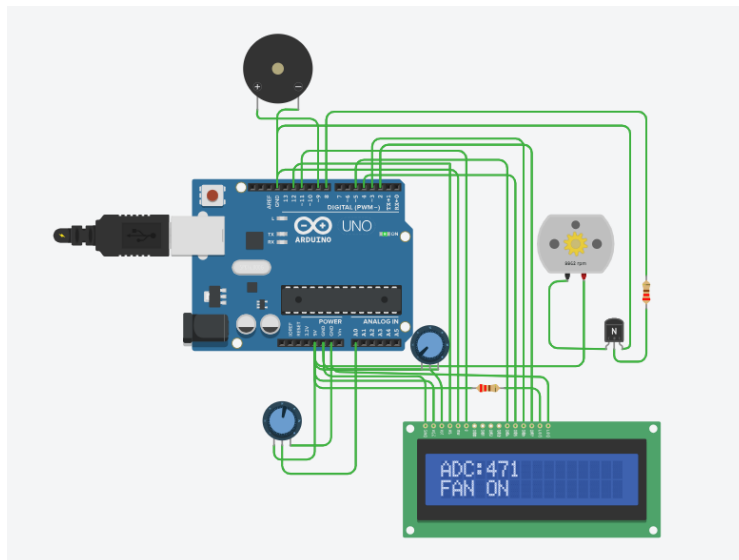


### Observations:

- LCD displayed "NORMAL".
- Cooling fan remained OFF.
- Buzzer remained OFF.
- System operated under safe conditions.

## 2. High Temperature Condition

When the potentiometer produced an ADC value between **301 and 700**, the system interpreted the temperature as approximately **38°C**.

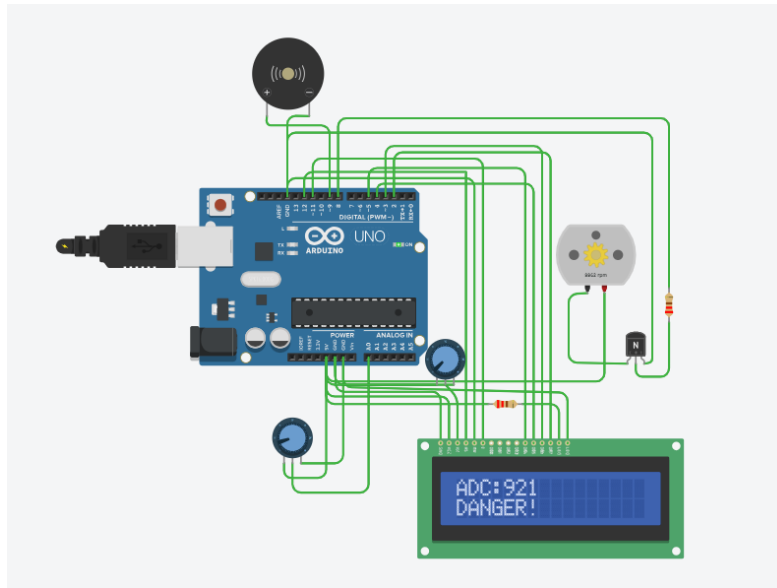


### Observations:

- LCD displayed "FAN ON" or "Cooling ON".
- Cooling fan was automatically activated.
- Buzzer remained OFF.
- Excess heat was addressed through automatic cooling.

### 3. Dangerous Temperature Condition

When the potentiometer produced an ADC value between **701 and 1023**, the system interpreted the temperature as approximately **40°C**.



### Observations:

- LCD displayed "DANGER!"
- Cooling fan remained ON.
- Buzzer was activated.
- Immediate warning was provided to indicate unsafe conditions.

### EXPERIMENTAL RESULTS:

| ADC Range  | Simulated Temperature | LCD Status | Fan | Buzzer |
|------------|-----------------------|------------|-----|--------|
| 0 – 300    | 35°C                  | NORMAL     | OFF | OFF    |
| 301 – 700  | 38°C                  | FAN ON     | ON  | OFF    |
| 701 – 1023 | 40°C                  | DANGER     | ON  | ON     |

## **DISCUSSION:**

The simulation results demonstrate that the proposed system can effectively monitor temperature conditions and provide automated responses without human intervention. The use of Arduino Uno enabled real-time monitoring and control of output devices such as the cooling fan and buzzer.

One of the major challenges encountered during the project was the limitation of temperature sensor simulation in Tinkercad. To overcome this issue, a potentiometer was utilized to generate varying analog input values that represent different temperature conditions. Although this approach does not measure actual temperature, it effectively demonstrates the logic and functionality of the proposed system.

The LCD display provided a user-friendly interface by continuously displaying the current operating condition. The automatic activation of the cooling fan and buzzer enhanced system reliability and demonstrated how embedded systems can improve patient safety in neonatal care environments.

Overall, the project successfully achieved its objectives of monitoring incubator temperature, providing automatic cooling control, and generating alerts during unsafe conditions. The system can be further enhanced by integrating real temperature sensors, humidity monitoring, IoT connectivity, and data logging capabilities for practical hospital applications.

## **CHALLENGES FACED:**

During the development of this project, several challenges were encountered:

1. Difficulty in obtaining proper temperature variation using available Tinkercad temperature sensors.
2. LCD display wiring issues, including contrast adjustment and backlight configuration.
3. Incorrect current flow through the LCD backlight causing simulation errors.
4. Integration of multiple components such as LCD, motor, transistor, buzzer, and potentiometer.
5. Calibration of threshold values to represent realistic temperature conditions.

To overcome these challenges, a potentiometer was used as a temperature simulator, and the LCD wiring was carefully corrected and tested.

## **ADVANTAGES:**

- Continuous temperature monitoring.
- Automatic cooling mechanism.
- Immediate danger alerts.
- Reduced manual intervention.
- Cost-effective implementation.
- Suitable for neonatal healthcare applications.

## **FUTURE SCOPE:**

- We can replace the potentiometer with a real temperature sensor such as LM35 or TMP36.
- Add humidity monitoring for complete incubator environmental control.
- Integrate IoT technology for remote monitoring by doctors and nurses.
- Implement data logging and cloud storage.
- Develop a mobile application for real-time alerts and monitoring.

## **CONCLUSION:**

The Smart Neonatal Incubator Monitoring and Temperature Control System demonstrates how embedded systems can be applied in biomedical engineering to improve infant healthcare. The system provides automated monitoring, cooling control, and emergency alerts. Although a potentiometer was used for simulation purposes in Tinkercad, the project can be easily upgraded with real temperature sensors for practical implementation. The proposed system enhances patient safety, reduces manual monitoring efforts, and contributes to improved neonatal care.

## **REFERENCES:**

### **1. Temperature and Humidity Monitoring System**

<https://github.com/AshutoshPant128/Temperature-and-Humidity-Monitoring-System-#temperature-and-humidity-monitoring-system>

### **2. Neonatal Incubator Temperature Monitor**

<https://github.com/Waxxy1404/Incubator-Temperature-Monitor>